

WEBINARS SERIES ON RAINWATER HARVESTING

6th September 2022

Module n°5:

**Ways to operate and
maintain RWH systems**



Module 5: Ways to operate and maintain RWH systems

**Session: Technical and socio-economic
considerations in RWH O&M**

Date: 6th September 2022 **Time:** 13 h 00 – 14 h 30 (GMT+2)

Register: Technical and socio-economic considerations in
RWH O&M

BACKGROUND

Water scarcity is a critical challenge in countries and regions with arid and semi-arid conditions and fast-growing water demands. This is mostly the case in all of the countries across the NENA Region, where food demand and food insecurity continue to grow, widening the water supply-demand gap in most countries. Furthermore, the anticipated impact of climate change in the NENA region is indicative towards an increased intensity of rainstorms and flash floods, while the total amount of annual precipitation rates shows declining trends. During the last 20 years, interest has been renewed in rainwater harvesting (RWH) as one of the most effective climate adaptation strategies to cope with water scarcity, thus reducing the pressure on the limited freshwater resources and increasing rainfed crop productivity.

This webinar series on Rainwater Harvesting aims to provide an overview of RWH applications and illustrates, with country examples, the diversity of approaches. It will try to unlock the potential of rainwater harvesting applications in the NENA region by highlighting the role of rainwater management in overcoming the current challenges in achieving water security. It will also shed the light on the need for innovations in economics, social consensus and supportive laws and regulations to ensure a widespread uptake of RWH technologies in the NENA region.

Main Conveners:

FAO/WEPS-NENA project & Water Scarcity Initiative (WSI) with: ICARDA, ICID and FAO inter-Regional Technical Platform on Water Scarcity (IRTP-WS)

Contributing Partners:

International Rainwater Harvesting Alliance

Contact Persons:

Domitille Vallée Domitille.Vallee@fao.org

Fahide Si Tahar Fahide.SiTahar@fao.org

Heba Al Hariry Heba.alhariry@fao.org

Simon Tanios Tanios.Simon@fao.org

Languages:

English and French with live translation

Recording:

<https://dgroups.org/fao/waterproductivity/rwh>

ABOUT THE MODULE 5

This module covers the operation and maintenance challenges of rainwater harvesting systems. It will discuss the approaches to O&M in the context of RWH systems and explore the extent to which O&M influences the sustainability of rainwater harvesting systems.



EXPECTED OUTCOMES FOR THE PARTICIPANTS

The webinar will achieve the following:

- 1 Examine the current practices, challenges and opportunities associated with the O&M of rainwater harvesting systems;
- 2 Bring attention to the importance of O&M for the sustainability of RWH systems and highlight the key requirements for the long-term performance of RWH systems.

SUGGESTED READINGS

- ➔ **Heijnen, H.** (2013). Enhancing economic resilience in North Eastern Brazil by harnessing rain. *Rainwater Harvesting Implementation Network*
http://www.abcmac.org.br/files/downloads/heijnen_rainwater_harvesting_in_semiard_brazil_compressed.pdf
- ➔ **Maddrell, S. R.** (2018). *Sand Dams: A Practical & Technical Manual*.
<https://accept-tchad.org/wp-content/uploads/2021/05/Sand-Dam-Manual-Version-3-ED-180614-print-lite.pdf>
- ➔ **Oweis, T., Prinz, D. & Hachum, A.** (2012). *Rainwater Harvesting for Agriculture in the Dry Areas*. CRC Press, 266 p.
- ➔ **Oweis T. Prinz, D. & Hachum** (2001). Water harvesting indigenous knowledge for the future of the drier environments
https://www.researchgate.net/publication/267131411_Water_harvesting_indigenous_knowledge_for_the_future_of_the_drier_environments_ICARDA_Aleppo_Syria_40_pp
- ➔ **Thomas, T. H., & Martinson, D. B.** (2007). *Roofwater Harvesting: A Handbook for Practitioners*. IRCWASH.
<https://www.ircwash.org/resources/roofwater-harvesting-handbook-practitioners>
- ➔ **Water from ponds, pans and dams. A manual on planning, design, construction and maintenance.** (2005) Technical handbook No 32. World Agroforestry Centre.
<http://apps.worldagroforestry.org/downloads/Publications/PDFS/MN13546.pdf>
- ➔ **WHO sanitary inspection formats: rainwater collection and storage (2020):**
https://cdn.who.int/media/docs/default-source/wash-documents/sanitary-inspection-packages/2-tfs-rainwater-collection-storage-d.pdf?sfvrsn=8dfb1bf8_6



CERTIFICATE OF ACHIEVEMENT

A digital certificate of achievement will be offered by FAO to participants who:

- Attend at least four webinars belonging to Module 1 to Module 4 and five webinars belonging to Module 5 to Module 10;
- Obtain a score of no less than 70 percent in each of two written assessments that include multiple-answer questions and reflect the content of the aforementioned modules.

The first assessment, covering the modules 1 to 4, was organized in August 2022. The link to the second assessment, covering the modules 5 to 10, will be sent by email at the end of the year to eligible participants who will be given two weeks to submit their answers. Respondents will have the possibility to make two attempts to answer the assessment questions, and only the highest score will be considered.

Important note: For participants who wish to obtain a certificate of achievement, to be able to track your attendance, you must register and join the webinars using the same e-mail address.

Speakers and moderator



Abdourahman Maki

is Land and Water officer at FAO sub-regional office for North Africa. He is the FAO focal point, for the Maghreb region, of the Regional Water Scarcity Initiative for Near East and North Africa region. He leads the planning, development and implementation of sub-regional programmes, projects and products on land and water management. Abdourahman Maki holds a PhD in Water Resources Development, a Master degree in Land and Water interaction and an Executive Master of Business Administration. He has over 20 years of working experience in the fields of hydrogeology, natural resources management, disaster risk reduction, and climate change in research, development and operation oriented institutions in various regions.



Theib Y. Oweis

Is a former director of the water, land, and ecosystems program of ICARDA and a world-leading specialist in water harvesting and management for agriculture, especially under conditions of water scarcity. He has a Ph.D. and MSc. degrees in Agriculture and Irrigation Engineering and spent over 35 years researching, developing, teaching, and publishing on water harvesting, supplemental irrigation, deficit irrigation, water productivity, salinity, and other areas associated with improving the productivity of water use in agriculture.



Han Heijnen

has worked in water supply and public health in Asia and Africa since 1979. In the early days of his career, water was often still sufficient although quality and service needed development. Since then, population growth, urbanization and climate change have started to affect the reliability and seasonality of water resources. Han Heijnen started developing rainwater solutions in rural water supply in Sri Lanka in 1995 as part of a World Bank investment program and has promoted rainwater harvesting in South Asia ever since. He has worked for the Governments of Nepal and Sri Lanka, for WHO, UNICEF, several NGOs and knowledge institutes. Since 2015, he is the President of the International Rainwater Harvesting Alliance, based in Geneva. He currently works with a Dutch NGO in promoting affordable domestic Calabash rainwater storage cisterns in over 10 countries in Africa.

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Date: 6th September, 2022

Time: 01:00 PM - 02:30 PM (GMT +2)

Time	Activity
01:00 – 01:05	Welcoming Remarks Introduction to the session objectives and speakers Moderator: Maki Abdourahman, Senior Land and Water Officer, FAO
01:05 – 01:30	Presentation on “Ensuring Sustainable Operation and Maintenance of RWH Systems in the MENA Region” By: Theib Oweis, Consultant in Water, Land and Ecosystems
01:30 – 01:55	Presentation on “Guidance and maintenance of RWH systems” By: Han Heijnen, President of the International Rainwater Harvesting Alliance
01:55 – 02:25	Open Discussions – Q&A's
02:25-02:30	Conclusion & Wrap-up



This activity is implemented within the regional project "Implementing the 2030 Agenda for water efficiency/productivity and water sustainability in NENA countries" under the umbrella of the Water Scarcity Initiative. This regional project is implemented by the Food and Agriculture Organization of the United Nations and funded by the Swedish International Development Cooperation Agency (SIDA).

Food and Agriculture Organization of the United Nations
Regional Office for the Near East and North Africa
www.fao.org/neareast
RNE-WEPS-NENA@fao.org

